

Announcements

- ▶ Any questions/issues?

- ▶ Schedule:

Today (New): Oceans and Ice

- ▶ Physics: coriolis, buoyancy, thermal expansion
- ▶ Climate/Weather: atmospheric and ocean currents, ENSO, Iceberg vs glacier melt, sea level

Next Week (Wed Lab): Oceans and Ice

Next Week (Tues/Thurs): Project Work.

This is the last lecture!

Today's Reading Quiz (3 min)

No reading for today, but find additional resources for these topics in:

- ▶ *Mathez*: Chap 3 — Ocean-Atmos Interactions (El Niño)
- ▶ *My notes*: Fluid statics (Buoyant Force, Floating)

both of which can be found on Blackboard under “Class Lecture Slides”

Final Poster Project

- ▶ Choose a topic that is interesting to you and make a poster to present (5min) to the class during our final exam period (Thurs, Dec 11, 10am–Noon).
- ▶ Work on your own or with a partner.
- ▶ Think of it as a Discussion/Lab/Lecture that you create, explaining some topic to yourself and us.
- ▶ **Talk to me by Thanksgiving (Tues Nov 25)** to verify your project topic (**that chat is worth one Discussion assignment**).
- ▶ In the last week of class you will have time (lecture and discussion) to work on your final poster project.

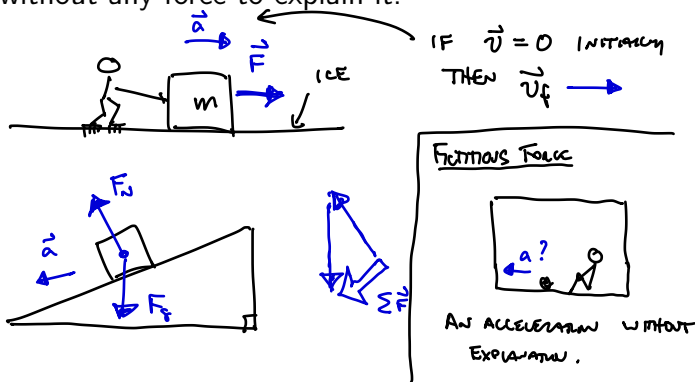
Newton's Laws and "Fictitious" Forces

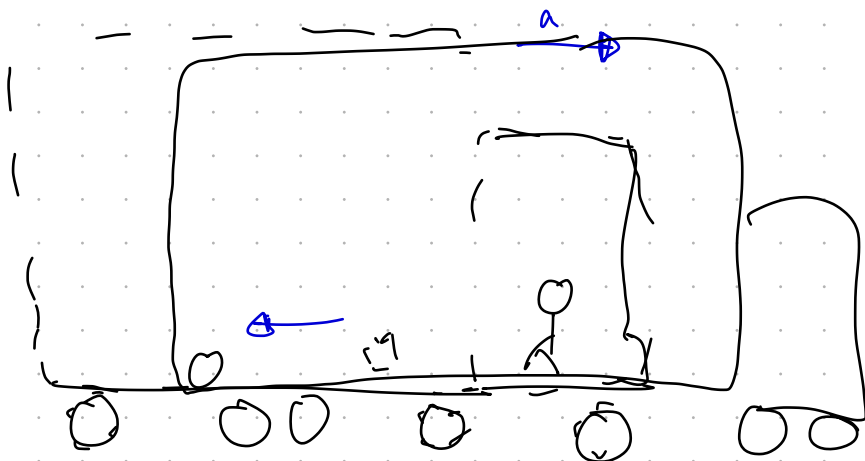
- Recall that Newton's Second Law tells us:

$$\sum \vec{F} = m \vec{a}$$

An object will *accelerate* (change its velocity vector) due to the sum of the forces acting on it.

- ▶ We say a force is “fictitious” if an object accelerates without any force to explain it.



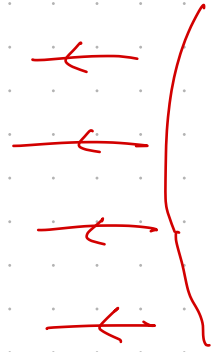


Coriolis Effect

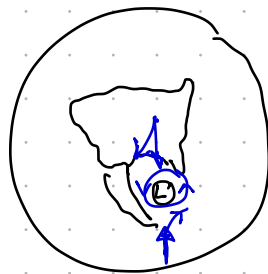
- ▶ The circulation of the atmosphere is due to fictitious forces arising from the rotation of the Earth. These are called **Coriolis** forces, or the “coriolis effect.”
- ▶ We can understand these dynamics by first **investigating the playground “merry-go-round” ride.**



- ▶ Net result: in the Northern Hemisphere flows “turn right”.



A diagram showing a cross-section of a dees (the semi-circular electrode) of a cyclotron. The dees is a semi-circle with a dashed line representing the particle's path. The path starts at the center, moves outwards, and then spirals inwards towards the center. The path is labeled "ELECTRON" and "CCW ROT." (Counter-Clockwise Rotation). The path is also labeled with V_{max} at the outer edge and V_{min} at the inner edge.

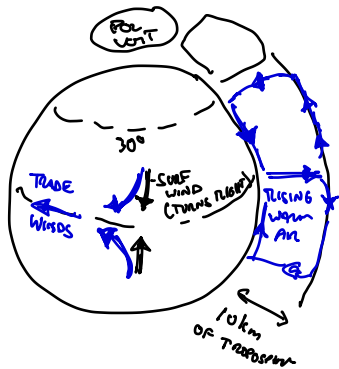
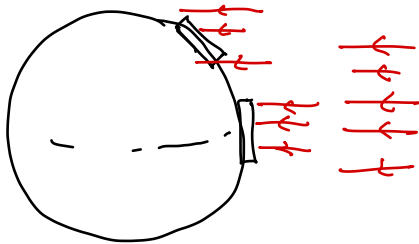


Flows in Northern Hemisphere Bend to their right.

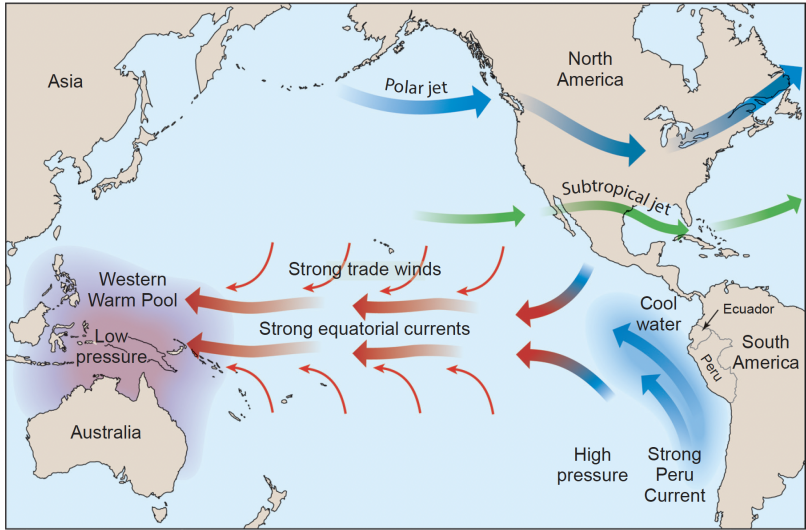
Coriolis Effect and Trade Winds

- Let's try to understand the origin of the "trade winds," that flow from East to West along the equator.

Solar insolation
is largest at
equator.

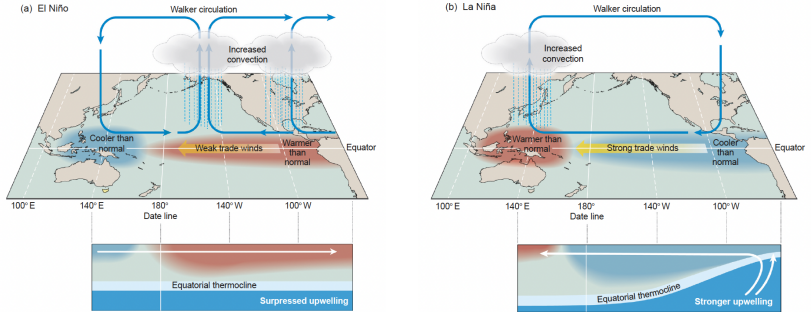


El Niño, La Niña, and ENSO



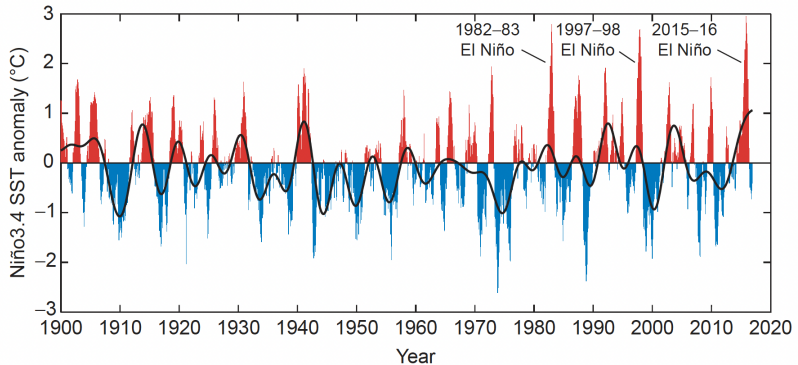
Mathez Figure 3.1: The neutral state of the tropical Pacific Ocean

El Niño, La Niña, and ENSO



Mathez Figure 3.4

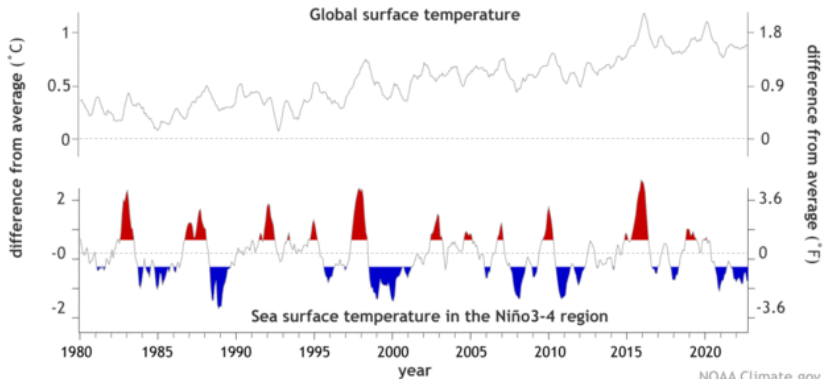
El Niño, La Niña, and ENSO



Mathez Figure 3.5

ENSO and Global Temperature

How global average surface temperature changes with **El Niño** and **La Niña**



NOAA Climate.gov
Mike McPhaden

[Climate.gov](https://climate.gov)

Buoyant Force, Floating, Ice

- ▶ We define the **buoyant force** on an object within a fluid as the sum of the pressure forces that it experiences.
- ▶ Since the displaced fluid is exactly held in equilibrium, the buoyant force is:

$$F_B = \text{weight of fluid displaced}$$

- ▶ We can use this to understand which objects float, and how much they are submerged.

